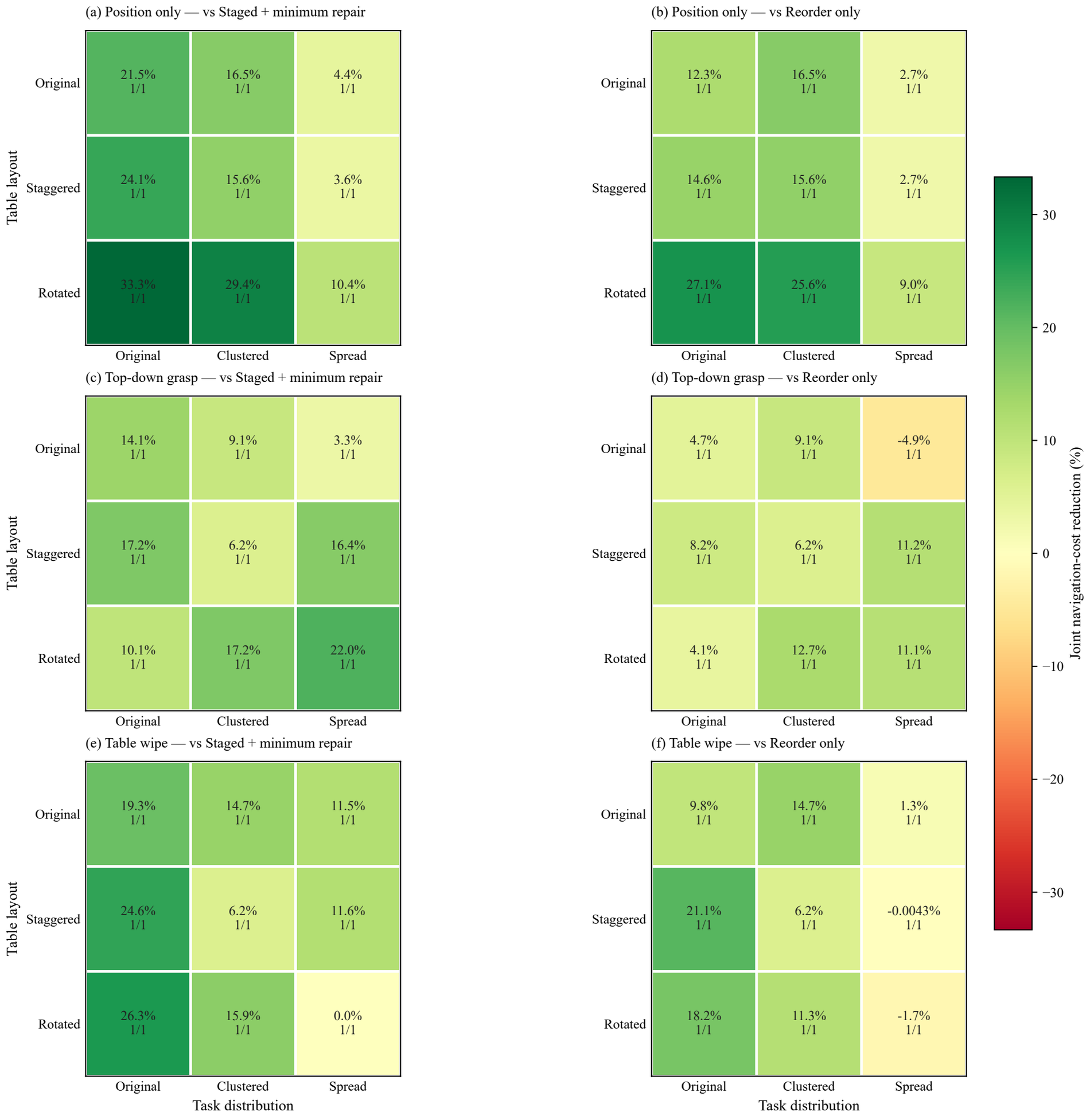
(a) Position only — vs Staged + minimum repair
(b) Position only — vs Reorder only
(c) Top-down grasp — vs Staged + minimum repair
(d) Top-down grasp — vs Reorder only
(e) Table wipe — vs Staged + minimum repair
(f) Table wipe — vs Reorder only
Table layout
Original
Staggered
Rotated
Task distribution
Original
Clustered
Spread
21.5% 1/1
16.5% 1/1
4.4% 1/1
24.1% 1/1
15.6% 1/1
3.6% 1/1
33.3% 1/1
29.4% 1/1
10.4% 1/1
12.3% 1/1
16.5% 1/1
2.7% 1/1
14.6% 1/1
15.6% 1/1
2.7% 1/1
27.1% 1/1
25.6% 1/1
9.0% 1/1
14.1% 1/1
9.1% 1/1
3.3% 1/1
17.2% 1/1
6.2% 1/1
16.4% 1/1
10.1% 1/1
17.2% 1/1
22.0% 1/1
4.7% 1/1
9.1% 1/1
-4.9% 1/1
8.2% 1/1
6.2% 1/1
11.2% 1/1
4.1% 1/1
12.7% 1/1
11.1% 1/1
19.3% 1/1
14.7% 1/1
11.5% 1/1
24.6% 1/1
6.2% 1/1
11.6% 1/1
26.3% 1/1
15.9% 1/1
0.0% 1/1
9.8% 1/1
14.7% 1/1
1.3% 1/1
21.1% 1/1
6.2% 1/1
-0.0043% 1/1
18.2% 1/1
11.3% 1/1
-1.7% 1/1
Joint navigation-cost reduction (%)
30
20
10
0
−10
−20
−30